URBAN VISION AI — SAFEBUILD

Community Access Review & Scenario Decision Document | SFIC 2026

Study Area: Kakkanad Urban Pilot Corridor (Kochi, Kerala, India)
Infrastructure Context: Kochi Metro Rail Phase II (JLN Stadium to Infopark via Kakkanad)
Scenario Label: HYPOTHETICAL_CONSTRUCTION_ACCESS_REVIEW_ON_REAL_ROADS
Analytical Mode: motor_vehicle | **Snapshot Date:** 2026-10-12
Evaluated Facility: Sunrise Hospital, Ernakulam (OPERATOR_ADDRESS_VERIFIED_ENTRANCE_PROVISIONAL)

1. Deterministic 5-State Scenario Attribution

Total Analyzed Origins: 6 | **Baseline Reachable:** 6 | **Under Construction Reachable:** 6
Worst Construction Detour: 354.9 m (Kakkanad West / Civil Station Environs) | **Average Detour:** 112.2 m
Proposal Algebra Invariant (G0+P - G0 == 0): SATISFIED (P = identity)

Origin Community	Households	G0 (m)	G0+C (m)	Detour (m)	Detour %	Mitigated (m)
DLF City / New Town Heights	1200	821.7	850.6	28.9	3.5%	850.6
Kollamkudimugal Residential Area	650	1579.7	1579.7	0.0	0.0%	1579.7
Canal Colony Settlement	420	2097.7	2097.7	0.0	0.0%	2097.7
Padamugal Residential Neighborhood	850	1831.9	2121.6	289.7	15.8%	2121.6
NGO Quarters Housing Colony	900	1405.6	1405.6	0.0	0.0%	1405.6
Kakkanad West / Civil Station Environs	500	1572.6	1927.5	354.9	22.6%	1927.5

2. Feasibility-Constrained Mitigation Optimization

Optimization Status: NO_VERIFIED_OPTION
Reviewer Note: All reviewed candidates lack verified statutory permission or physical clearance. System refuses to present unverified options as actionable advice.

Candidate	Feasibility Status	Permission Status	Detour Reduction	Conflict
Single-Lane Managed Contraflow & Signal Priority	GRAPH_ONLY_UNVERIFIED	UNVERIFIED_PENDING_TRAFFIC_POLICE_REVIEW	0.0 m	PEDESTRIAN
Civil Lane Road Formal Secondary Diversion	GRAPH_ONLY_UNVERIFIED	LOCAL_MUNICIPALITY_NOTIFIED_NOT_SANCTIONED	0.0 m	NONE
Direct Infopark Boundary Compound Cut-Through	REJECTED_UNFEASIBLE	PERMISSION_REJECTED_PRIVATE_CAMPUS	0.0 m	PEDESTRIAN

3. Scientific Limitations & Audit Trail

• **Non-Emergency Claim:** Detour distances are deterministic graph computations and do not represent observed ambulance response times, traffic congestion, or live travel speeds.
• **Unverified Permissions:** No reviewed candidate has been granted statutory police or municipal road opening permissions. The system deliberately refrains from marking them implementable.
• **Data Source:** OpenStreetMap contributors (ODbL). Snapshot downloaded 2026-10-12 across 19,582 nodes with complete referential topology.